

SPECTRA SCHEDULER

Adaptive spectrum scanning
A simulation-first software prototype

Problem Statement ID: SIH26055

Problem Statement Title

**Smart Scan strategy for
Electronic Warfare**

PS Category: Software | Organisation: DRDO

Theme: [Confirm exact portal category]

Team ID: _____

Team Name (registered): _____



Spectra Scheduler: learn where and when to listen

Proposed solution: a feedback-driven scheduler for a receiver that can observe only a small part of the spectrum at once.

Detailed explanation of the proposed solution

Simulate changing emitters.
Observe noisy hits and misses.
Associate pulses into tracks.
Choose the next band and dwell.

How it addresses the problem

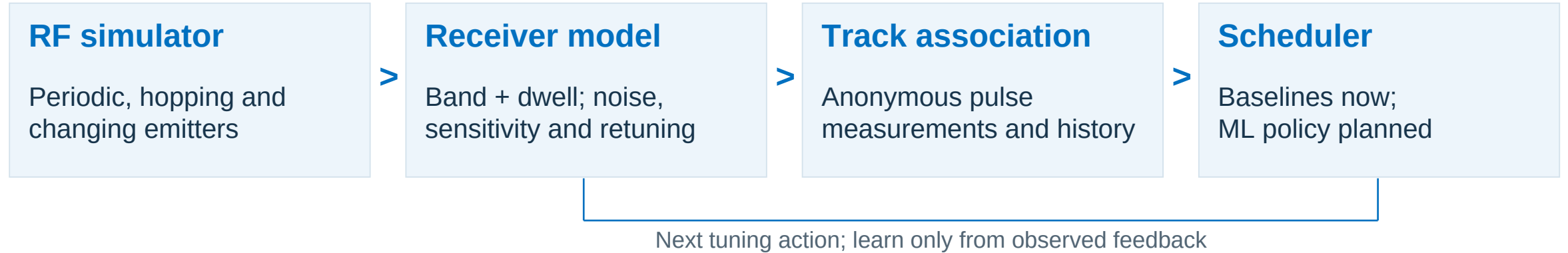
Update from new observations.
Revisit likely signal activity.
Retain broad-band discovery.
Include receiver retuning cost.

Innovation and uniqueness of the solution

Combine track confidence, change response and coverage.
Test each part against baselines.
Keep decisions inspectable.

OBSERVE > ASSOCIATE > SCHEDULE > UPDATE

Methodology and process for implementation



Technologies to be used

Python + NumPy; simple package modules.
HDF5 / h5py for bounded data batches (next).
scikit-learn baseline; PyTorch if justified (next).
Rust / PyO3 only for profiled bottlenecks (later).

Data and learning plan

TSRD: arrival time, frequency, width, angle, amplitude.
Keep train / validation / test files separate.
First: pulse-clustering baseline on a small subset.
Next: contextual-bandit scheduler trained on simulator hits/misses; test on held-out scenarios.

Truth labels go to evaluation only at inference. | Reports first; API and GUI later. [1–3]

Analysis of the feasibility of the idea

123

automated tests passing

13

comparison strategies

5

seeded scenario presets

Working now: simulation, tracking, parallel runs, custom scenarios and JSON/CSV reports.

Potential challenges and risks	Strategies for overcoming these challenges
Large pulse files exceed available memory	Read bounded HDF5 batches; profile before native code.
Similar signatures cause false joins or splits	Measure pairwise F1 and V-measure; test noisy overlaps.
Simulation bias or changing emitter behaviour	Hold out scenarios; vary noise and modes; keep baselines.
Recorded scans cannot reveal missed alternatives	Evaluate scheduling in simulation; replay data for association.

Next: dataset evaluation, learned policy, reward/prediction metrics and a periodic-scan reference.
Simulation evidence only; hardware integration and field validation remain outside this prototype.

Potential impact on the target audience

For electronic-support researchers, receiver developers and student RF labs:
a repeatable way to compare scan policies before considering hardware integration.

Technical benefits

Aim for faster signal discovery.
Balance revisits with coverage.
Explain misses and track errors.

Economic benefits

Start on existing computers.
Test software before RF trials.
Reuse scenarios and reports.

Research and education

Reproducible, inspectable tests.
Public synthetic benchmarks.
No live RF transmission needed.

How success will be measured

Receiver: detection / false-alarm probability and sensitivity. Scheduling: interception rate, reward/cost, prediction accuracy and intercept-time error. Association: V-measure and pairwise F1.

Details / links of the reference and research work

- 1 Gunn et al. — The Turing Synthetic Radar Dataset**
Benchmark design, scan/stare receiver modes and pulse-association evaluation.
arxiv.org/abs/2602.03856
- 2 Alan Turing Institute — TSRD dataset and companion code**
Five pulse features; file-level splits; HDF5 loading and reference evaluation.
[Hugging Face dataset](#) | [Turing Deinterleaving Challenge](#)
- 3 Gunn et al. — Transformer-based deep metric learning**
Research direction for learned pulse embeddings; not implemented in this prototype.
arxiv.org/abs/2503.13476
- 4 SIH26055 — Smart Scan strategy for Electronic Warfare**
Problem requirements: receiver simulation, feedback learning and interception metrics.
[SIH portal](#) | [Accessible statement copy \(community mirror\)](#)

Literature informs the approach; published benchmark scores are not presented as our results.